

**M.D. GENERAL MEDICINE
CASE PROFORMA CENTRAL NERVOUS SYSTEM**

Name:
Age/Sex:
Occupation:
Address:
Handedness:

HISTORY

PRESENTING COMPLAINTS:

Sequence of events in the patients own words

HISTORY OF PRESENTING ILLNESS:

1. WEAKNESS OF LIMBS

- a. Inability to use the limbs
- b. Onset - Sudden/insidious including the time of occurrence
- c. Duration
- d. Severity paresis or paralysis
- e. Progession If chronic which limb affected first, time interval between one limb and another limb and whether the disease is static, progressive, worsening or improving
- f. Which side and which limb is affected
 1. Upper limbs
 - i) Proximal combing the hair, taking the food to the mouth
 - ii) Distal buttoning the shirt, mixing the food
 2. Lower limbs
 - i) Proximal squatting and getting up from the squatting position, climbing stairs up and down
 - ii) Distal slipping of chappals, tripping of toes
- g. Whether the neck is affected difficulty in lifting the head off the pillows
- h. Whether the trunk is affected Rolling over the bed, getting up from the bed
- i. Difficulty in breathing
- j. Whether there is any diurnal variation in the weakness of the muscles
- k. Whether the limbs are flaccid and loose or stiff and hard

2. SPINOMOTOR SYSTEM

- a. Wasting/Thinning of muscles
- b. Pain/ fatigue/ muscle cramps
- c. Fasciculations/muscle twitchings
- d. Involuntary movements chorea, athetosis, tremors, hemiballismus

3. SENSORY SYSTEM

- a. sensory deficit able to feel clothes, feeling hot and cold water while bathing
- b. numbness/pins and needle sensations, sensation of walking on cotton wool

- c. root pain/neck pain/back pain
- d. band like sensation/girdle like sensation
- e. unsteadiness on closing the eyes/washbasin sign/able to walk in the dark
- f. Lhermitte's symptom

4. HIGHER MENTAL FUNCTIONS

- a. Loss of consciousness/alteration in sensorium duration, complete/partial, aura, associated with fits, bladder/bowel disturbances, recovery and frequency of occurrence/fits
- b. Speech disturbances aphasia, slurring of speech(cerebellum), slowness of speech(extrapyramidal system)/memory/sleep disturbances
- c. Delusions/ Hallucinations/emotional disturbances

5. CRANIAL NERVES

- I alteration in smell
- II blurring of vision/diminished vision, differentiate colours, night blindness
- III, IV, VI drooping of eyelids/ double vision/ able to move eyes in all directions
- V having sensation over the face, able to chew food
- VII able to close the eyes and lips, deviation of angle of mouth, drooling of saliva, able to feel taste of objects
- VIII hard of hearing, tinnitus, vertigo
- IX, X difficulty in speech, nasal twang to speech, regurgitation of feeds, difficulty in swallowing
- XI move the neck in all directions, lift the shoulder
- XII able to roll the tongue and push the food backwards

6. CEREBELLUM

- a. spilling of food while taking to the mouth, clumsiness of hands
- b. unsteadiness while walking and swaying to sides
- c. ability to negotiate narrow pathways

7. AUTONOMIC NERVOUS SYSTEM

- a. bladder able to feel bladder fullness, initiate micturition, feel the passage of urine, able to hold urine midstream, able to completely evacuate the bladder without any dribbling inability to pass urine, catheterisation
- b. bowel constipation, loss of control leading to involuntary defecation
- c. giddiness on getting up in the morning
- d. sweating disturbances, palpitation

8. MENINGES

Fever, vomiting, stiffness of neck

8. RELEVANT NEGATIVE HISTORY:

- a. trauma, fall from height, lifting heavy objects on the head/back/ spinal anaesthesia
- b. headache
- c. vomiting, diarrhoea
- d. calf tenderness
- e. chest pain
- f. drug intake antihypertensives, oral contraceptives, anticoagulants, vaccination, any injections

- j. bed sores, dark coloured urine

PAST HISTORY:

- a. Similar illness in the past
- b. TIA/RIND
- c. Comorbid illness like Diabetes, Hypertension, coronary artery disease, thyroidal illness, Tuberculosis, HIV, malignancy
- d. Trauma
- e. fever
- f. Drug intake and vaccination for dog bite
- g. Previous medical illness/surgeries
- h. Developmental history

PERSONAL HISTORY:

- a. Married
- b. Veg or non-veg
- c. Sleep pattern
- d. Smoking
- e. Alcohol
- f. Drug abuse
- g. Exposure

FAMILY HISTORY:

- a. Family tree
- b. Consanguinity
- c. Any other affected family members

MENSTRUAL HISTORY AND OBSTETRIC HISTORY FEMALES:

- a. Menarche
- b. Regular/Irregular/LMP
- c. No. of children
- d. Full term/Preterm/Vaginal/LSCS
- e. Obstetric complications
- f. Menopause

TREATMENT HISTORY:

- a. Previous Treatment
- b. Present Treatment

SUMMARY:

- 1. Onset acute/subacute/insidious
- 2. Progression static, remittent, intermittent, rapidly progressive, slowly progressive, improving
- 3. Disease Paraplegia/quadriplegia/hemiplegia/multiple cranial nerve palsy/others
- 4. Structures involved cortex/brain stem/ long tracts/ anterior horn cells/ anterior or posterior nerve roots/ peripheral nerves/ myoneural junction/ muscles/ cerebellum/extrapyrmidal/autonomic

5. Levels Single/Multiple
6. Etiology Trauma/vascular/infective/inflammatory/degenerative/demyelinating/malignancy

PHYSICAL EXAMINATION

GENERAL EXAMINATION:

Patient is conscious/not
 Oriented/not
 Comfortable/not
 Co-operative/not
 Built
 Nourishment
 Febrile/afebrile
 Palor
 Icterus
 Cyanosis central/peripheral
 Clubbing bilateral/unilateral, pandigital/limited
 Pedal edema bilateral/unilateral, painless/painful, pitting (soft/hard)/non-pitting, regional
 lymphadenopathy
 Significant lymphadenopathy

SHORT NECK(Height/Neck Ratio), SCARS, BURNS, DESTRUCTION OF TERMINAL PHALANX, DECUBITUS ULCERS, TROPHIC ULCERS

RIGHT OR LEFT HANDED INDIVIDUAL, CATHETERISED/NOT

NEUROCUTANEOUS MARKERS

1. TUBEROUS SCLEROSIS adenoma sebaceum, shagreen patches, ash leaf macules
2. NEUROFIBROMATOSIS cafe u lait spots, neurofibromas, periungual fibromas, axillary freckling
3. VON-HIPPEL LINDAU SYNDROME conjunctival hemangioma
4. STURGE WEBER SYNDROME port wine stain(capillary hemangioma)
5. ATAXIA TELANGIECTASIA telangiectasias
6. SPINA BIFIDA tuft of hair
7. PHAKAMOTOSSES birth marks, nevus
8. FRIEDREICH'S ATAXIA pes cavus

VITAL SIGNS

PULSE: rate, rhythm, volume, character, felt in all peripheral pulses/not, radial/radial/radiofemoral delay, apex pulse deficit, condition of vessel wall

BLOOD PRESSURE: _____mm of Hg measured in the _____ Upper limb with the patient in sitting position and _____mm of Hg in the standing position. Look for Postural Hypotension

RESPIRATORY RATE: ____/min, regular/not, type-thoracoabdominal/abdominothoracic

TEMPERATURE: _____ F measured in the Axilla

NEUROLOGICAL EXAMINATION

1. HIGHER MENTAL FUNCTIONS:

- a. Consciousness
- b. Orientation to time, place and person
- c. Speech and language aphasia, dysarthria, dysphonia
- d. Memory immediate-retention and recall, recent and remote
- e. Delusions, hallucinations
- f. Emotional lability
- g. MMSE score
 - I. Orientation
 1. date, day, month, season, year (5)
 2. floor, hospital. District, state, country (5)
 - II. Registration
Name three objects taking one second for each object. Ask him to repeat the same.
Repeat till he remembers (3)
 - III. Attention and Calculation
Serial 7s 5 times (5)
 - IV. Recall
Recall the three objects (3)
 - V. Language
 1. Name 2 objects (2)
 2. Repeat a sentence (1)
 3. Follow a 3 stage command (3)
 4. Reading close your eyes(1)
 5. Writing a sentence (1)
 6. Copy a design (1)

2. CRANIAL NERVES

CRANIAL NERVE	TEST	RIGHT	LEFT
I	Sense of smell i) Coffee ii) Asafoetida	+/- +/-	+/- +/-
II	i) Visual acuity Rosenbaum Chart ii) Field of vision Confrontation test iii) Colour vision Ishihara chart iv) Fundus	20/20 Normal Normal Normal	20/20 Normal Normal Normal
III, IV, VI	i) Extra-ocular movements ii) Pupil Size iii) Direct Light Reflex iv) Consensual Light Reflex v) Accommodation Reflex vi) Ptosis	full mm Present Present Present Absent	full mm Present Present Present Absent

	vii) Nystagmus viii) Horner's syndrome	Absent No	Absent No
V	i) Sensory -over face and buccal mucosa ii) Motor masseter, temporalis, pterygoids iii) Reflex a. Corneal Reflex b. Conjunctival Reflex c. Jaw jerk	Normal Normal Present Present Present	Normal Normal Present Present Present
VII	i) Motor nasolabial fold hyperacusis occipitofrontalis orbicularis oculi orbicularis oris buccinator platysma ii) Sensory Taste of anterior 2/3rds of tongue(salt/sweet) Sensation over tragus iii) Reflex Corneal Conjunctival iv) Secretomotor Moistness of the eyes/tongue and buccal mucosa	Present Absent Good Good Good Good Good Normal Normal Present Present Normal	Present Absent Good Good Good Good Good Normal Normal Present Present Normal
VIII	i) Rinne's Test ii) Weber's Test iii) Nystagmus	Positive Not lateralised Absent	Positive Absent
IX, X	i) Uvula, Palatal arches, and movements ii) Gag reflex iii) Palatal reflex	Centrally placed and symmetrical Present Present	 Present Present
XI	i) trapezius ii) sternocleidomastoid	Good Good	Good Good
XII	i) Tone ii) Wasting iii) Fibrillation iv) Tongue Protrusion to the midline and either side	Normal No No Normal	Normal No No Normal

3. MOTOR SYSTEM

TEST	RIGHT	LEFT
<u>I BULK</u> a. Inspection	Normal	Normal

b. Palpation c. Measurements Upper limb 10cm above and below acromion Lower limb 18 cm above and 10 cm below tibial tubercle	Normal cms cms	Normal cms cms
<u>II TONE</u> a. Upper limbs b. Lower limbs	Normal/rigidity/spasticity	Normal/rigidity/spasticity
<u>III POWER</u> a. Neck muscles b. Upper limbs i) Shoulder Flexion-Extension Lateral Rotation-Medial Rotation Abduction -Adduction ii) Elbow Flexion-Extension iii) Wrist Dorsi flexion-Palmar flexion Abduction-Adduction Pronation-Supination iv) small muscles of hand v) Hand grip c. Lower limbs i) Hip Flexion-Extension Abduction-Adduction Lateral Rotation-Medial Rotation ii) Knee Flexion-Extension iii) Ankle Dorsi flexion-Plantar flexion Inversion-Eversion iv) Small muscles of foot d. Trunk muscles e. Beevors sign	Good 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 Good Good 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 Good Absent	Good 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 Good Good 5/5 5/5 5/5 5/5 5/5 5/5 5/5 5/5 Good Absent
<u>IV REFLEXES</u> <u>A. SUPERFICIAL REFLEXES</u> 1. Corneal 2. Conjunctival 3. Pharyngeal Reflex 4. Palatal Reflex 5. Abdominal Reflex 6. Cremasteric Reflex 7. Perianal Reflex	Present Present Present Present Present Present Present	Present Present Present Present Present Present Present

8. Plantar Reflex	Flexor	Flexor
<u>B. DEEP TENDON REFLEXES</u>		
1. Jaw jerk	Present	
2. Trapezius jerk	Present	Present
3. Pectoralis jerk	Present	Present
4. Biceps jerk	Present	Present
5. Triceps jerk	Present	Present
6. Supinator jerk	Present	Present
7. Finger flexion reflex	Present	Present
8. Knee jerk	Present	Present
9. Ankle jerk	Present	Present
10. Clonus	Absent	Absent
	Absent	Absent
<u>C. PRIMITIVE REFLEXES</u>		
<u>D. OTHERS</u>	Absent	Absent
Hoffmans sign	Absent	Absent
Wartenbergs sign	Absent	Absent
Rossalimos sign	Absent	Absent
<u>V COORDINATION</u>		
TESTED ALONG WITH THE CEREBELLUM		
<u>VI GAIT</u>		
<u>VII INVOLUNTARY MOVEMENTS</u>		
A Athetosis, Asterexis		
B Ballismus		
C Chorea		
D Dystonia		
E Essential tremors		
F Fasciculations		
M - Myoclonus		

4. SENSORY SYSTEM

TEST	RIGHT	LEFT
<u>I SPINOTHALAMIC</u>		
1. Crude touch	Normal	Normal
2. Pain	Normal	Normal
3. Temperature	Normal	Normal
<u>II POSTERIOR COLUMN</u>		
1. Fine touch	Normal	Normal
2. Vibration	Normal	Normal
3. Position sense	Normal	Normal
4. Rombergs sign	Absent	Absent
<u>III CORTICAL</u>		
1. Two point discrimination	Normal	Normal
2. Tactile localisation	Normal	Normal
3. Graphaesthesia	Normal	Normal
4. Stereognosis	Normal	Normal

5. CEREBELLAR SIGNS

1. Titubation
2. Truncal ataxia/gait ataxia/ stance ataxia
3. Nystagmus
4. Dysarthria
5. Hypotonia
6. Rebound phenomenon
7. Intention tremor
8. Pendular knee jerk
9. Tandem Walking
10. Coordination
 - a. Upper Limbs Finger Nose test, Finger Finger Nose test, Drawing a circle, Putting a dot in the centre of the circle
 - b. Lower Limbs Heel Knee test, Drawing a circle
 - c. Dysdiadokokinesia

6. AUTONOMIC NERVOUS SYSTEM

Postural Hypotension
Resting tachycardia
Abnormal sweating

7. SIGNS OF MENINGEAL IRRITATION

Neck stiffness
Kernigs sign
Brudzinskis sign

8. EXAMINATION OF THE SPINE AND CRANIUM

SPINE bony deformities, bruit, gibbus, tenderness
CRANIUM bony abnormalities, bruit

9. PERIPHERAL NERVES

Thickened nerves
Trophic ulcers
Wrist drop
Foot drop

10. CAROTID PULSES AND BRUIT

EXAMINATION OF OTHER SYSTEMS

CARDIOVASCULAR SYSTEM:

JVP, Apex normally placed, no Palpable P2, Heart sounds normal, No thrills/murmurs

RESPIRATORY SYSTEM:

Chest symmetrical, No paradoxical movements, Normal vesicular breath sounds heard,
No abnormal/added sound

ABDOMEN:

Abdomen is soft, No organomegaly, No ascites, Per-rectal examination

DIAGNOSIS**INVESTIGATIONS****TREATMENT**
